Biologically Interpretable Graph Generation for Brain Graph Analysis

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***Abstract*—Brain graph analysis using graph machine learning has received considerable attention. In these studies, brain graphs are typically generated from functional magnetic resonance imaging by defining brain regions as nodes and estimating their connections using temporal correlations between regional signals. Although widely used, temporal correlation methods assume that connections between brain regions remain stable and estimate connectivity by averaging co-activation states across the entire time period. This assumption overlooks the biological fact that these states are inherently dynamic, with some persisting only for very short durations. Averaging across such dynamic states can obscure transient co-activation patterns and lead to inaccurate graph representation. To address this limitation, we propose a biologically interpretable brain graph generation model that moves beyond traditional temporal correlation. Grounded in the biological nature of brain connectivity, our approach estimates connections by mining transient co-activation patterns, thereby producing more reliable graph structures. The model comprises two key components: a co-activation pattern miner that detects transient co-activations between brain regions through frame-wise analysis of fMRI data, and a brain graph generator that leverages the detected co-activations to estimate inter-regional connections and generate brain graphs. The generated graph structures are compatible with a wide range of graph analysis models and fMRI datasets without requiring architectural or dataset-specific modifications, demonstrating the flexibility and practical applicability of the proposed method across diverse experimental settings.**

***Index Terms*—Deep learning, brain disorder prediction, brain graph generation, fMRI.**

I. Introduction

In recent years, graph machine learning–based brain graph analysis has been increasingly applied to the diagnosis of brain disorders. In light of the well-documented association between brain disorders and functional connectivity abnormalities [1], these studies primarily focus on analyzing the connection patterns within brain graphs to facilitate diagnosis. Typically, a brain graph is generated by parcellating the brain into regions of interest (ROIs) using a brain atlas, where each ROI serves as a node. Functional connections between nodes, represented as edges, are commonly estimated from the temporal correlations (e.g., Pearson correlation coefficient) of functional magnetic resonance imaging (fMRI) time series between ROIs.

While most brain graph analysis approaches generate graphs in this manner and focus on developing models to extract informative patterns for diagnosis [2], [3], [4], [5], [6], [7], [8], an important issue is often overlooked: ***whether temporal correlation–based connection estimation can effectively represent inter-regional brain connectivity?*** Biologically, connections between brain regions are inherently dynamic [9], [10]. As shown in Fig. 1 (a), the fMRI signals between ROI-3 and ROI-4 alternate between positive and negative correlation states, with some states persisting only for very short time scales. These transient states play a critical role in predicting brain disorders [11], [12]. However, temporal correlation averages these dynamic states over the entire period, yielding a single correlation strength that reflects only overall connectivity. This averaging risks cancelling out opposing transient neural interactions, causing the loss of important information and ultimately reducing the reliability of connectivity estimation. To address this limitation, several studies have proposed generating dynamic brain graphs. These methods typically apply a sliding window to the fMRI time series, segmenting it into shorter intervals so that connectivity can be estimated within each segment [13], [14], [15], [16]. Estimating connectivity within shorter windows reduces the risk that opposing connection states co-occur within the same period. However, as shown in Fig. 1 (b), this approach cannot fully resolve the issue when such states persist for durations shorter than the window length, and thus may still overlook critical neural interactions [11]. A straightforward solution is to further reduce the window size to improve the identification of transient states. However, smaller windows increase the variance of correlation estimates, which undermines the reliability of the generated graphs. Moreover, they produce a larger number of graph snapshots, substantially raising computational cost and reducing efficiency. In summary, temporal correlation–based connection estimation cannot fully capture transient interactions between brain regions.

To address this issue, we draw inspiration from co-activation pattern–based connection estimation [11], [12], a method that has been widely applied in neurological research but is rarely employed in machine learning approaches for brain graph generation. In this method, fMRI signals at each time point are treated as independent instances of neural activity, and frame-wise analysis is performed to detect co-activations of brain regions, which are then used to estimate their functional connections. Here, co-activations are defined as simultaneous high-amplitude fMRI signals, and brain regions repeatedly engaged in the same co-activation potentially constitute a co-activation pattern. As illustrated in Fig. 1 (c), co-activation pattern analysis enables the effective capture of transient connection states and prevents critical connections from being overlooked.

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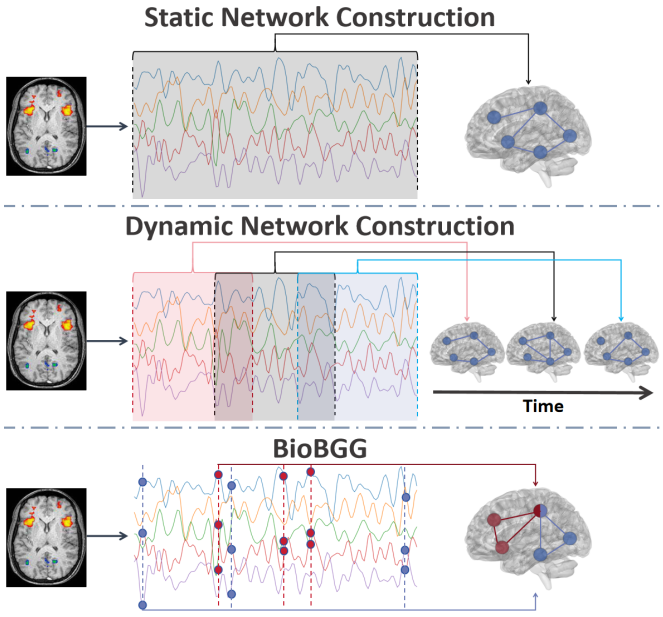


Fig. 1. Visualization of different brain graph generation methods. From top to bottom, the figure illustrates the static, dynamic, and proposed BioBGG approaches. In the signal sequence plots, red and blue boxes denote positive and negative correlations, respectively. In the brain graphs, red edges indicate positive connections, while blue edges indicate negative connections.

To leverage co-activation patterns for brain graph generation, two main challenges arise: (1) **Learning critical (task-relevant) co-activation patterns**. In the brain, co-activation patterns are abundant, and attempting to capture all of them is computationally inefficient and may reduce model interpretability. Previous studies [2], [4] have shown that only a small subset of task-relevant connections exhibit group-level discrepancies, which are critical for disorder prediction. Learning co-activation patterns associated with these connections can substantially reduce computational cost while enhancing diagnostic accuracy. (2) **Detecting these patterns in individuals**. The presence or absence of the learned critical co-activation patterns in individuals provides essential cues for prediction, as certain patterns are characteristic of healthy subjects while others are associated with brain disorders. Because these patterns may appear only in limited frames of fMRI data, detecting them accurately is particularly important for ensuring the reliability of brain graphs.

Based on the above analysis, we propose the **Bio**logically Interpretable **B**rain **G**raph **G**eneration Model (BioBGG) for brain disorder analysis, a deep learning model that leverages transient co-activations detected from fMRI signals to generate brain graphs. Specifically, to address the challenges of learning critical co-activation patterns and detecting them in individuals, we develop a critical co-activation pattern miner module. In this module, a series of trainable vectors are initialized as candidate patterns for learning critical co-activation patterns. These candidate patterns participate in brain graph generation and are optimized with respect to downstream objectives to achieve task-aware adaptation. Together with the proposed critical ROI enhancement regularization, each candidate pattern eventually represents a specific critical co-activation pattern. Detection of the presence of these candidate patterns in individuals is achieved by computing a frame-wise spatial similarity score between signal patterns and the candidate patterns. Since higher scores indicate a greater likelihood of a candidate pattern's occurrence at a given time point, temporal max pooling is applied to extract a series of local maxima, whose average yields a pattern existence score that reflects the likelihood of each candidate pattern's presence in the brain. Finally, to transform the detected candidate patterns into a brain graph, we introduce a brain graph generator module. Guided by the candidate patterns and their corresponding existence scores, this module estimates the connections among the ROIs involved in each pattern, generating biologically meaningful brain graphs that can be seamlessly integrated into brain disorder analysis models. The main contributions of this work are summarized as follows:

- We identify the limitations of temporal correlation–based brain graph generation in capturing transient neural interactions, and propose a co-activation pattern–based model to address this issue, introducing a novel perspective for brain graph generation.
- We perform task-aware brain graph generation and identify a series of task-relevant co-activation patterns. The results demonstrate that these patterns align well with disorder-related biomarkers reported in previous studies, underscoring the interpretability of the proposed model.
- We conduct extensive experiments to evaluate the proposed model. The results confirm its superior performance and demonstrate compatibility with diverse brain graph analysis models and fMRI datasets without requiring architectural or dataset-specific modifications.

II. Related Work

A. Static Brain Graph-Based Methods

Brain graphs derived from fMRI have been widely applied in studies of brain disorders [17], [18]. Among various graph generation strategies, static brain graphs remain the most widely adopted and have demonstrated competitive performance in recent research. These methods assume that brain connectivity is stationary over time and estimate inter-regional connections using the temporal correlation of the entire fMRI time series. Based on the architecture of the employed graph analysis models, these methods can be broadly categorized into three groups: autoencoder-based, graph neural network (GNN)-based, and transformer-based.

Autoencoder-based models [19], [20] typically flatten the Pearson correlation coefficient-based connection matrix into a one-dimensional feature vector for brain disorder prediction. For example, Heinsfeld et al. [19] used the upper-triangular elements of the correlation matrix as input features and randomly masked a small subset of them to train an autoencoder for reconstruction. This strategy encourages the model to learn noise-tolerant embeddings, thereby improving prediction accuracy. Building on this idea, Almuqhim et al. [20] introduced a sparsity constraint on model parameters to focus on critical connections, further enhancing performance. Despite these competitive results, such models are often criticized

for ignoring the topological structure of brain graphs and providing limited interpretability.

In contrast, GNN-based methods [21], [2], [4] generate an adjacency matrix from thresholded correlations and represent each ROI by its connection profile, defined as the corresponding row of the correlation matrix [17], [18]. Using these representations, Li et al. [21] proposed an ROI-aware embedding kernel to emphasize the distinct identity of nodes within the brain graph. Cui et al. [2] and Li et al. [4] developed GNN models that suppress irrelevant connections, showing that emphasizing task-specific connections that are consistent across individuals yields better performance than focusing on varying connections across different individuals. This approach produces sparser brain graphs, which not only improve predictive performance but also enhance interpretability. In addition, Wen et al. [22] proposed using multiple thresholds to obtain multiview representations of the same graph, further boosting model performance. Collectively, these approaches improve diagnostic accuracy and facilitate the identification of critical connections associated with specific neurological conditions, improving the interpretability of the model.

To further improve performance, Kan et al. [3] introduced transformer-based architectures into brain graph analysis by applying self-attention mechanisms to model relationships between nodes. Building on this, Bannadabhavi et al. [5] incorporated prior knowledge of brain functional communities to guide node feature updates, while Yu et al. [23] employed a random walk–based mechanism to capture long-range dependencies. These methods consistently improve prediction accuracy, demonstrating the effectiveness of transformer architectures in brain graph analysis.

All of the aforementioned methods, regardless of the graph analysis model employed, rely on temporal correlation–based brain graphs, which fail to capture transient neural interactions. Consequently, this limitation inevitably constrains the performance of these models.

B. Dynamic Brain Graph-Based Methods

In addition to static brain graphs, a growing body of research has investigated dynamic brain graphs to capture short-duration connectivity patterns and temporally varying configurations. Most of these approaches segment the fMRI time series into multiple overlapping windows using a sliding-window technique, then compute temporal correlation within each window to generate a sequence of time-resolved graphs [13], [24], [25], [26]. For example, Kim et al. [13] employ this strategy to generate time-resolved graphs, apply GIN [27] to aggregate spatial information and produce graph-level embeddings, and then utilize a self-attention mechanism to capture temporal dependencies, thereby generating spatial-temporal embeddings for downstream tasks. Similarly, Yin et al. [24] and Lin et al. [28] utilize GNNs for spatial embedding generation but model temporal dependencies with LSTMs [29]. In contrast, Gadgil et al. [30] also define spatial connections using temporal correlations but explicitly link nodes representing the same brain regions across consecutive snapshots, enabling direct spatio-temporal convolution. Liu et al. [16] proposed a nodal feature–based contrastive learning model that adaptively generates temporal links between nodes across snapshots, allowing more flexible modeling of temporal evolution compared with fixed connection strategies. Wang et al. [31] also performed node-level convolution but applied self-attention across all nodes from all snapshots to aggregate global information, which they argue better captures long-term dependencies. In another line of work, Zhang et al. [32] introduced a kernel self-attention mechanism as an alternative to correlation for measuring connectivity, enabling the modeling of asynchronous co-activation from time-lagged signals and yielding improved performance.

Despite these advances, dynamic methods still rely on temporal correlations of signals over a given period to measure connections, which limits their ability to capture transient co-activation patterns that are critical for brain disorder diagnosis. Furthermore, the need to generate and analyze multiple graph snapshots substantially increases computational cost. These challenges underscore the necessity of developing alternative strategies for brain graph generation.

III. Problem Definition

In this work, we aim to generate a brain graph $\mathbf{A} \in \mathbb{R}^{N\times N}$ from fMRI signal sequences $\mathcal{S} \in \mathbb{R}^{N\times L}$, where N denotes the number of ROIs and L denotes the number of time points. To capture transient co-activations, we perform a frame-wise analysis of the fMRI data, treating the signals as a sequence of isolated neural activity instances, denoted as $\mathcal{S} = \{\mathbf{S}_t\}_{t=1}^{L}$. Each instance $\mathbf{S}_t = \{s_{i,t}\}_{i=1}^{N}$ corresponds to the t-th column vector of $\mathcal{S}$, where t denotes the time point and $s_{i,t}$ represents the signal value of the i-th ROI at time t. We term $\mathbf{S}_t$ a signal pattern. The goal is to extract co-activation patterns from these signal patterns and leverage them to construct a brain graph $\mathbf{A}$. Let $f_\theta(\cdot)$ denote the graph generation function and $f_\delta(\cdot)$ the graph classification function. The overall brain graph classification process can then be formulated as $y' = f_\delta\left(f_\theta\left(\mathcal{S}\right)\right)$.

IV. Method

To overcome the limitations of temporal correlation–based brain graph generation, we propose BioBGG, a brain graph generation model comprising two key components: a co-activation pattern miner and a brain graph generator. As illustrated in Fig. 2, the co-activation pattern miner is designed to learn critical co-activation patterns and identify their existence in the brain. It employs a set of candidate patterns to perform frame-wise analysis and compute similarity scores, which are then aggregated via temporal max pooling to capture local maxima, representing the likelihood of existence of specific co-activation patterns in the brain. Unlike traditional methods that rely on full-sequence signal analysis, this approach focuses on a small subset of informative time points, thereby enabling the detection of transient neural co-activations. Guided by the identified patterns, the brain graph generator estimates connection strengths between the corresponding ROIs and generates the resulting brain graph. The following subsections provide a detailed description of each component.

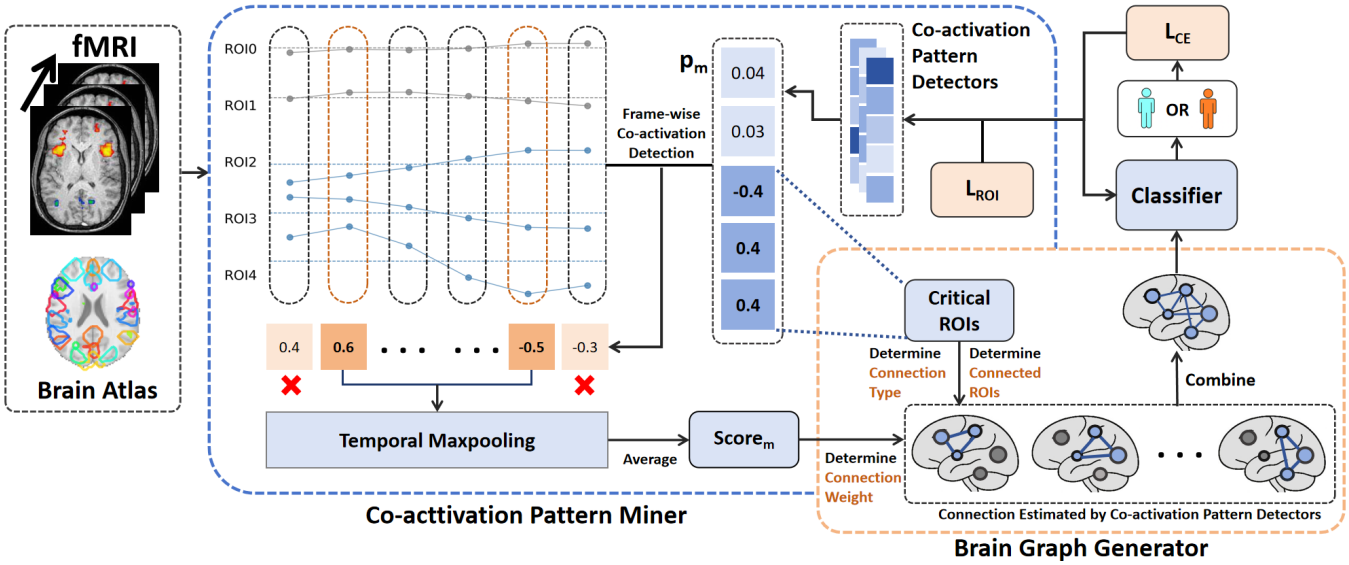


Fig. 2. The overall structure of the proposed BioBGG model. The left component illustrates the co-activation pattern miner, where candidate patterns are trained to represent critical co-activation patterns and used to detect similar signal patterns within fMRI sequences. The detection results are expressed as similarity scores, which are subsequently aggregated through a temporal max-pooling layer to produce pattern existence scores that indicate the likelihood of each candidate pattern's presence. The right component depicts the brain graph generator, which leverages the detected patterns and their corresponding existence scores to estimate inter-ROI connections and construct a brain graph compatible with standard brain graph classifiers. The entire model is jointly optimized with respect to the classification objective, thereby ensuring task-aware graph generation.

A. Co-activation Pattern Miner

The co-activation pattern miner is responsible for learning critical co-activation patterns and detecting their presence in the fMRI signals, which are then used to support the estimation of connection strengths in brain graph generation. The operation of this module is illustrated as follows.

1) fMRI Signal Normalization: Inspired by recent clinical research on co-activation pattern–based fMRI signal analysis [11], [12], we treat signals at each time point as isolated neural activities, which we define as a signal pattern. Within a signal pattern, if certain ROIs exhibit high activation levels, we regard them as potential participants in a co-activation pattern. However, because brain regions inherently differ in their baseline activation levels, which may bias the analysis, we apply ROI-wise z-score normalization to the fMRI signals before feeding them into the model, defined as:

$$\hat{s}_{i,t} = \frac{s_{i,t} - \mu_i}{\sigma_i}, \tag{1}$$

where $\mu_i = \frac{1}{L}\sum_{t=1}^{L} s_{i,t}$ denotes the mean signal of the i-th ROI across all time points, and $\sigma_i = \sqrt{\frac{1}{L}\sum_{t=1}^{L}(s_{i,t} - \mu_i)^2}$ denotes the corresponding standard deviation. We denote the collection of normalized signals as $\hat{\mathcal{S}}$, and the normalized signal pattern at time point t as $\hat{\mathbf{S}}_t$, which will serve as the basis for subsequent analysis.

2) Learning Critical Co-activation Patterns: To learn critical co-activation patterns, we initialize a set of trainable vectors using Xavier uniform initialization [33]. These vectors serve as candidate patterns, denoted as $\mathcal{P} = \{\mathbf{P}_m\}_{m=1}^{n}$, where n denotes the number of candidate patterns and $\mathbf{P}_m = \{p_{m,i}\}_{i=1}^{N}$. Here, $p_{m,i}$ represents the participation level of the i-th ROI in the m-th candidate pattern. The candidate patterns participate in detecting similar signal patterns within the fMRI data and are trained in an end-to-end manner to adapt to the downstream classification objective. Consequently, higher participation levels indicate that the corresponding ROI receives greater emphasis during candidate-pattern matching, guiding the model to focus on co-activation patterns that are more informative for the downstream diagnosis task. In addition to the classification objective, we introduce a critical ROI enhancement loss to further refine the candidate patterns for this task. The details of this loss are provided in Section IV-A4, following the co-activation pattern detection process, as its motivation and implementation are directly tied to that process.

3) Co-activation Pattern Detection: The co-activation pattern detection process includes two steps, which are frame-wise similarity score calculation and temporal max pooling-based co-activation patterns existence measurement.

Frame-wise similarity score calculation. As illustrated in Fig. 2, the likelihood of the existence of a co-activation pattern at a given time point is quantified by the similarity score between the corresponding candidate pattern and the signal pattern at that time. Specifically, the similarity score between the m-th candidate pattern and the signal pattern at time point t is computed using absolute cosine similarity, defined as:

$$\mathrm{Sim}(\mathbf{P}_m, \hat{\mathbf{S}}_t) = \left| \frac{\mathbf{P}_m^{\top}\hat{\mathbf{S}}_t}{\|\mathbf{P}_m\|_2 \left\|\hat{\mathbf{S}}_t\right\|_2} \right|, \tag{2}$$

where $\top$ denotes the transpose operator, $\|\cdot\|_2$ represents the L2 norm, and $|\cdot|$ denotes the absolute value. By calculating similarity scores across all time points, we perform a frame-wise analysis of fMRI signals, facilitating the capture of transient neural interactions.

One potential concern with this approach is its computational cost. Theoretically, if each candidate pattern focuses on a co-activation pattern involving k ROIs, there are $\binom{N}{k} = \frac{N!}{k!(N-k)!}$ possible combinations of co-activation patterns among N ROIs, making exhaustive modeling computationally infeasible. However, prior studies by Cui et al. [2] and Li et al. [4] have shown that only a small subset of connections is relevant to downstream tasks, and these task-relevant connections remain consistent across subjects, depending primarily on the nature of the task. Since our model performs task-aware brain graph generation, this observation supports the feasibility of employing a limited number of candidate patterns that consistently focus on a small subset of co-activation patterns, while still capturing sufficient information for accurate brain graph generation.

Temporal max pooling-based co-activation patterns existence measurement. Generally, the presence of a co-activation pattern at a single time point indicates its occurrence in the brain. Consequently, the maximum similarity score can serve as strong evidence of a co-activation pattern's existence. However, relying solely on a single time point may make the model vulnerable to coincidental co-activations, thereby compromising stability. Furthermore, fMRI signal sequence lengths often vary across subjects. Selecting the maximum or top-k similarity score from sequences of different lengths introduces inconsistencies in the expected value of the selected score. This creates an undesirable dependency between the similarity score distribution and sequence length, making it an unreliable indicator of co-activation pattern presence.

To address this limitation, and inspired by prior observations that brain activity exhibits recurring co-activation patterns [34], we apply a temporal max-pooling operation to capture these recurring patterns, resulting in a series of local maxima from fixed-length windows of the similarity score sequence. The pooled maxima are subsequently averaged to yield a more stable and robust measure, referred to as the pattern existence score. Formally, this is expressed as:

$$\text{Score}_m = \frac{1}{\lfloor \frac{L}{w} \rfloor} \sum_{j=1}^{\lfloor \frac{L}{w} \rfloor} \max \left(\left\{ \text{Sim}\left(\mathbf{P}_m, \hat{\mathbf{S}}_t \right) \;\middle|\; t \in T_j \right\} \right), \quad (3)$$

where Score_m denotes the pattern existence score corresponding to the m-th candidate pattern, and T_j is the set of all time points within the j-th max pooling window. Both the window size and stride are set to w. The floor operator $\lfloor \cdot \rfloor$ ensures that only complete windows are considered, thereby maintaining consistency in the number of time points included in each window. This temporal max pooling strategy enables the reliable detection of transient co-activation patterns while enhancing robustness against coincidental activations.

4) Critical ROI Enhancement Loss: For each candidate pattern, we expect only a few elements to have substantially larger values than the others, thereby emphasizing specific ROIs while allowing the remaining ones to be disregarded. The number of these emphasized ROIs is denoted as k, which we refer to as the kernel size in the following discussion. Designing co-activation patterns with a small kernel size serves two primary purposes.

First, focusing on too many ROIs reduces the accuracy of detecting co-activation patterns between specific pairs of regions. In practice, signal patterns may not perfectly align with a given candidate pattern. When more than two ROIs are included, the similarity score alone cannot identify which specific ROIs deviate from the candidate pattern, introducing ambiguity in the connections between these ROIs and thereby reducing interpretability. Second, the max pooling operation selects the highest similarity scores, which correspond to signal patterns that are most consistent with the candidate patterns. When candidate patterns are designed with a strong contrast between large values for critical ROIs and smaller values for others, the model is encouraged to favor signal patterns with a similar distribution. As a result, frames in which the critical ROIs exhibit higher amplitudes relative to the rest are more likely to be selected. This aligns with the definition of a co-activation (the simultaneous occurrence of high-amplitude signals), thereby facilitating the identification of more reliable co-activations. To explicitly enforce this property, we propose a critical ROI enhancement loss that constrains candidate patterns to emphasize only a small subset of ROIs. The loss is formally defined as:

$$\mathcal{L}_{\text{ROI}} = \frac{1}{n} \sum_{m=1}^{n} \left(1 - \sqrt{\frac{\sum_{i \in \text{ROI}_m^k} (p_{m,i}^2)}{\sum_{i=1}^{N} (p_{m,i}^2)}} \right)^2, \quad (4)$$

where ROI_m^k represents the set of k critical ROIs of the m-th candidate pattern, and $p_{m,i}$ denotes the i-th element of the m-th candidate pattern. This loss encourages the sum of squared magnitudes of the k elements corresponding to the critical ROIs to approximate that of the entire vector, thereby amplifying the contrast between the emphasized and non-emphasized ROIs. The critical ROIs are defined by selecting the indices of the top-k elements in the m-th candidate pattern:

$$\text{ROI}_m^k = \text{TopK}\left(\{|p_{m,i}|\}_{i=1}^{N}, k \right), \quad (5)$$

where TopK returns the indices of the k elements with the largest magnitude values in $\{p_{m,i}\}_{i=1}^{N}$.

B. Brain Graph Generator

This component aims to transform the observed co-activation patterns into a brain graph. For each candidate pattern, the sign of its elements indicates the type of connection state being evaluated. For example, as illustrated in Fig. 2, the candidate pattern $\mathbf{P}_m$ evaluates whether there is a negative connection between ROI-2 and ROI-3, as well as between ROI-2 and ROI-4, and whether there is a positive connection between ROI-3 and ROI-4. The corresponding pattern existence score then determines whether such connections are present in the brain. To facilitate this process, we first simplify each candidate pattern by retaining only the sign of its elements, as defined below:

$$p'_{m,i} = \begin{cases} 1, & \text{if } i \in \text{ROI}_m^k \quad \text{and} \quad p_{m,i} > 0 \\ -1, & \text{if } i \in \text{ROI}_m^k \quad \text{and} \quad p_{m,i} < 0 \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

where $p'_{m,i}$ denotes the sign of the i-th element of the m-th candidate pattern. Notably, elements corresponding to non-critical ROIs are set to zero, as their values are suppressed by the critical ROI enhancement loss and have minimal influence on the similarity scores. Then, the edge weight between ROI_i and ROI_j is computed as:

$$e_{i,j} = \frac{\sum_{m=1}^{n} p'_{m,i} p'_{m,j} \text{Score}_m}{\sum_{m=1}^{n} |p'_{m,i} p'_{m,j}| + \epsilon}. \tag{7}$$

The denominator in this equation serves to normalize the connection values that are evaluated multiple times across different candidate patterns, while ϵ is a small positive constant used to avoid division by zero. If we denote the matrix formed by the simplified candidate patterns as $\mathbf{P}'$, the diagonal matrix formed by all pattern existence scores as $\mathbf{D}_{\text{score}} = \text{diag}\{[\text{Score}_1, \text{Score}_2, \ldots, \text{Score}_n]\}$, and let $\mathbf{1}$ denote an all-ones matrix, the above computation can be rewritten in matrix form to obtain the final brain graph $\mathbf{A}$ as follows:

$$\mathbf{A} = \frac{\mathbf{P}' \mathbf{D}_{\text{score}} \mathbf{P}'^{\top}}{|\mathbf{P}' \mathbf{P}'^{\top}| + \epsilon \mathbf{1}}, \tag{8}$$

where the division is performed element-wise.

Algorithm 1 BioBGG-based brain disorder prediction

Require: fMRI data $\mathcal{S}$, number of candidate patterns n, kernel size k, window size w, loss weight α, analysis model $f_\delta(\cdot)$, learning rate η
Ensure: Trained candidate patterns $\mathcal{P}$ and predicted label y'

Initialize candidate patterns $\mathcal{P}$ and model parameters δ.
Normalize data: $\hat{\mathcal{S}} \leftarrow \text{Norm}(\mathcal{S})$ (Eq. (1)).
for each training iteration **do**
  for all $\hat{\mathbf{S}} \in \hat{\mathcal{S}}$ **do**
    Compute similarity scores: $\mathbf{Sims} \leftarrow \text{Sim}(\mathcal{P}, \hat{\mathbf{S}})$ (Eq. (2)).
    Compute pattern existence scores: $\mathbf{Scores} \leftarrow \text{TempMaxPool}(\mathbf{Sims}, w)$ (Eq. (3)).
    Generate simplified candidate patterns: $\mathcal{P}' \leftarrow \text{Simplify}(\mathcal{P})$ (Eq. (6)).
    Generate graph: $\mathbf{A} \leftarrow \text{GraphGen}(\mathcal{P}', \mathbf{Scores})$ (Eq. (8)).
    Make prediction: $y' \leftarrow f_\delta(\mathbf{A})$.
    Compute $\mathcal{L}_{\text{ROI}}$ (Eq. (4)).
    Compute $\mathcal{L}_{\text{CE}}$ (Eq. (9)).
    Update model parameters: $\delta \leftarrow \delta - \eta \nabla_\delta \mathcal{L}_{\text{CE}}$.
    Update candidate patterns: $\mathcal{P} \leftarrow \mathcal{P} - \eta \nabla_{\mathcal{P}} (\mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_{\text{ROI}})$.
  end for
end for
Return $\mathcal{P}$, y'

C. Training Objective

The brain graph $\mathbf{A}$ is directly used as input to brain graph analysis models to generate the prediction. During training, both the proposed brain graph generation model and the employed analysis model are jointly optimized for the downstream task using the cross-entropy loss, defined as:

$$\mathcal{L}_{\text{CE}} = -\left(y \log(y') + (1-y) \log(1-y')\right), \tag{9}$$

where $y \in \{0, 1\}$ is the ground-truth label and y' is the predicted probability of assigning the input sample to the positive class. Incorporating the proposed constraints on candidate patterns, the final loss function is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_{\text{ROI}}, \tag{10}$$

where α is a hyperparameter used to adjust the weight of $\mathcal{L}_{\text{ROI}}$. Finally, the overall process of BioBGG-based brain disorder prediction is presented in Algorithm 1.

D. Computational Complexity Analysis

The time complexity of the proposed brain graph generation method is determined by the number of ROIs N, the number of candidate patterns n, and the signal sequence length L, yielding a complexity of $\mathcal{O}(N \cdot n \cdot L)$ (with $n = 4N$ in this study). This reflects a computational cost that is comparable to the widely used temporal correlation-based static brain graph generation, which exhibits a complexity of $\mathcal{O}(N^2 \cdot L)$ and is commonly recognized for its computational efficiency. A detailed analysis of the time cost is provided in Section V-H.

V. Experiment

In this section, we validate the effectiveness of the proposed brain graph generation model by addressing the following questions:

Q1. How does BioBGG perform when applied to primary brain graph analysis models, and does it outperform other state-of-the-art brain graph generation methods?

Q2. What are the contributions of the key components of BioBGG?

Q3. How do hyperparameters affect the performance of BioBGG?

Q4. How computationally efficient is BioBGG compared with other brain graph generation methods?

Q5. How interpretable is BioBGG, and how is it more reliable than temporal correlation–based methods?

A. Dataset

Three publicly available datasets that are widely used in open scientific research are included in this study: the ABIDE [35] dataset, the ADHD [36] dataset, and the HCP [37] dataset. The ABIDE and ADHD datasets are used for Autism Spectrum Disorder (ASD) and Attention Deficit Hyperactivity Disorder (ADHD) diagnosis, respectively, while the HCP dataset is used for gender classification.

For the ABIDE dataset, we included 871 samples that passed quality control, among which 403 individuals were diagnosed with ASD, and 468 individuals were healthy controls (HC). All samples underwent preprocessing using the Configurable Pipeline for the Analysis of Connectomes [38], which includes slice time correction, motion correction, skull stripping, nuisance signal regression, band-pass filtering (0.01-0.1 Hz), and registration. The ADHD dataset consists of 936

samples, including 356 individuals diagnosed with ADHD and 580 HC. All samples in this dataset were processed using the Athena pipeline [36]. The fMRI data from both the ABIDE and ADHD datasets were further partitioned into 200 ROIs based on the CC200 brain atlas [39]. The HCP dataset includes 1,008 subjects, with 469 males and 539 females. fMRI data from this dataset were preprocessed following the methodology described in [40] and partitioned into 200 ROIs with group Independent Component Analysis. This research is about fundamental data science research, involving graph analysis algorithms from a data mining perspective and does not involve any treatments and interventions for human health or disease.

B. Experimental Settings

In this study, all models are implemented using the PyTorch framework and trained on an NVIDIA RTX 4080 Super graphics card with 16 GB of memory.[1] There are four hyperparameters that are set manually: the number of candidate patterns (n), the kernel size (k), the window size (w) of the max pooling layer, and the weight (α) for $\mathcal{L}_{\text{ROI}}$. These were set to 800, 3, 25, and 0.25, after a candidate selection process. All model parameters were optimized using the Adam optimizer [41] with a learning rate of 0.0001 and a weight decay of 0.00001. The training process was conducted over 100 epochs with a batch size of 32. To reduce variance in the results, all reported outcomes are based on the average of five iterations of five-fold cross-validation, each initialized with a different random seed. In each fold, the data were randomly split into training, validation, and test sets in a 7:1:2 ratio.

C. Evaluation Metrics

We assess the proposed method using four widely adopted evaluation metrics [2], [3], [23]:

- **Accuracy**: Proportion of correctly classified samples.
$$\textbf{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$
- **Sensitivity**: Proportion of correctly identified positives.
$$\textbf{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
- **Specificity**: Proportion of correctly identified negatives.
$$\textbf{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$
- **AUC (Area Under the ROC Curve)**: Quantifies the model's capacity to distinguish between classes, with higher values reflecting stronger discriminative ability.

Here, **TP**, **TN**, **FP**, and **FN** denote true positives, true negatives, false positives, and false negatives, respectively.

D. Baselines

We select nine baseline models to evaluate the effectiveness of BioBGG. Based on the type of brain graph used, these baselines are grouped into three categories. The first category comprises models that utilize static brain graphs, where connections are estimated using full-sequence temporal correlations. We include three representative models with different architectures from this category:

- **AE** [19]: an autoencoder-based model that learns noise-tolerant embeddings, producing robust results.
- **GCN** [42]: a widely used graph learning model that has demonstrated strong performance in brain graph analysis.
- **BNT** [3]: a transformer-based model specifically designed for brain graph analysis, which has served as the backbone of numerous brain disorder prediction studies.

The second category, referred to as advanced static brain graphs, includes three models. These methods either use modified correlation-based graphs as input or generate brain graphs through alternative approaches beyond correlation estimation. The baselines in this category are:

- **IBGNN** [2] and **IGS** [4]: two GNN-based models that sparsify correlation-based brain graphs to emphasize task-relevant connections and improve interpretability.
- **FBNETGEN** [43]: a framework integrates deep learning–based brain graph generation with the downstream classifier. By jointly optimizing graph generation and the classifier, it enables task-aware brain graph generation.

The third category consists of dynamic brain graph–based methods, and we include three representative models:

- **STAGIN** [13]: a model that generates spatial graph embeddings using GCN and captures temporal dependencies with self-attention.
- **STIGR** [44]: a model that aggregates local information with GNN and applies attention across all nodes to model node-level spatio-temporal awareness.
- **ACIFBN** [32]: a model that employs a kernel-based attention mechanism to estimate connections, enabling capturing both synchronous and asynchronous co-activations.

E. Performance Comparison (Q1)

We demonstrate the superiority of the proposed model through two sets of experiments. In the first set, we compare traditional correlation-based static graphs with the proposed BioBGG. To eliminate the influence of the analysis model itself, we replace only the brain graph while keeping the analysis model unchanged when comparing with these baselines. As shown in Tab. I, across all three datasets, the models achieve performance improvements on nearly all metrics when correlation-based graphs are replaced with BioBGG, highlighting both the effectiveness of the proposed brain graph generation method and its adaptability to widely used analysis models. The only exception is a drop in sensitivity on the ADHD dataset. Given the substantial increase in specificity, this behavior appears to reflect a trade-off between identifying more ADHD patients and reducing false-positive diagnoses. Based on the reliability analysis in Section V-I, BioBGG tends to capture more consistent discriminative features. Given the heterogeneity across ADHD patients and the noise present in the ADHD dataset, some less reliable or subject-specific discriminative features may receive less emphasis. Consequently, subjects exhibiting weaker or less consistent disease-related connectivity patterns may be classified as healthy controls, leading to lower sensitivity but higher specificity. Importantly,

[1]The code is available at https://anonymous.4open.science/r/Anonymize123

TABLE I
PERFORMANCE COMPARISON (MEAN ± STD, %) BETWEEN CORRELATION-BASED STATIC GRAPHS AND BIOBGG. "CORR." DENOTES TEMPORAL CORRELATION–BASED BRAIN GRAPHS. THE HIGHEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Baseline	Variant	ABIDE				ADHD				HCP			
		Accuracy	AUC	Sensitivity	Specificity	Accuracy	AUC	Sensitivity	Specificity	Accuracy	AUC	Sensitivity	Specificity
AE	Corr. (Original)	66.80±1.01	72.95±4.25	72.57±1.29	59.86±0.67	62.80±2.05	64.57±3.77	**41.41±1.35**	70.36±1.58	91.48±0.36	92.65±0.24	91.20±0.64	94.09±0.37
	+BioBGG	**71.23±1.58**	**77.56±0.83**	**76.52±1.31**	**65.49±1.82**	**65.65±1.27**	**67.81±1.03**	39.55±1.95	**79.91±1.61**	**92.32±0.48**	**95.36±0.60**	**91.51±0.55**	**94.24±0.43**
	(Improv.)	+4.43	+4.61	+3.95	+5.63	+2.85	+3.24	-1.86	+9.55	+0.84	+2.71	+0.31	+0.15
GCN	Corr. (Original)	63.41±1.19	66.50±1.28	66.45±0.69	63.25±1.48	60.29±1.20	61.56±1.33	**42.59±7.80**	72.73±1.79	82.50±7.83	89.72±3.66	72.68±7.02	90.48±3.15
	+BioBGG	**69.48±0.57**	**76.65±0.97**	**76.25±1.38**	**63.91±0.62**	**64.52±1.04**	**66.61±1.37**	41.85±1.73	**77.56±1.42**	**90.58±1.35**	**95.67±0.52**	**89.48±1.45**	**92.13±1.48**
	(Improv.)	+6.07	+10.15	+9.80	+0.66	+4.23	+5.05	-0.74	+4.83	+8.08	+5.95	+16.80	+1.65
BNT	Corr. (Original)	68.05±1.01	72.90±0.60	72.98±1.11	62.11±1.93	64.02±0.87	64.91±1.19	**43.72±3.46**	75.61±2.01	91.70±0.84	95.36±0.46	90.84±1.04	93.48±2.13
	+BioBGG	**70.15±1.27**	**74.40±1.88**	**73.38±1.76**	**68.04±2.81**	**65.88±1.25**	**66.64±1.13**	41.69±3.68	**81.95±3.88**	**92.64±0.72**	**95.80±0.58**	**92.10±1.30**	**94.03±1.07**
	(Improv.)	+2.10	+1.50	0.40	+5.93	+1.86	+1.73	-2.03	+6.34	+0.94	+0.44	+1.26	+0.55

TABLE II
PERFORMANCE COMPARISON (MEAN ± STD, %) ACROSS BASELINES FOR DIFFERENT BRAIN GRAPH GENERATION METHODS. THE HIGHEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Network Type	Baseline	ABIDE				ADHD				HCP			
		Accuracy	AUC	Sensitivity	Specificity	Accuracy	AUC	Sensitivity	Specificity	Accuracy	AUC	Sensitivity	Specificity
Advanced Static	IBGNN	63.68±1.77	67.15±1.11	64.12±2.65	57.85±1.47	60.82±1.87	63.51±1.33	**42.05±1.70**	73.08±1.52	83.41±4.32	89.15±4.97	77.38±9.01	89.85±6.14
	IGS	65.62±1.84	67.55±2.54	64.12±1.47	66.88±2.58	62.45±1.54	64.27±1.75	41.24±5.08	73.02±3.30	86.63±5.42	92.17±4.78	85.47±3.02	89.86±4.54
	FBNETGEN	66.27±1.54	68.75±2.11	65.78±1.35	**67.56±2.05**	61.20±1.30	64.13±1.95	39.80±4.29	73.32±2.24	91.14±0.74	95.20±0.55	90.52±1.34	93.00±1.41
Dynamic	STAGIN	56.57±1.01	55.17±1.28	60.91±4.24	47.86±4.06	54.25±1.25	53.18±1.34	38.20±4.04	66.02±3.85	88.46±4.55	93.24±1.66	77.99±15.73	92.04±7.48
	STIGR	61.32±1.84	63.05±2.42	60.32±1.49	62.78±2.58	60.22±1.47	62.27±1.45	38.28±4.18	72.01±3.30	84.56±2.32	88.47±3.42	83.07±3.02	88.61±4.02
	ACIFBN	62.72±1.54	64.33±2.44	61.12±1.57	65.31±1.88	65.21±1.54	66.24±1.65	41.14±5.08	73.12±3.28	85.37±4.22	90.03±3.58	83.37±3.22	90.13±4.02
BioBGG + AE		**71.23±1.58**	**77.56±0.83**	**76.52±1.31**	65.49±1.82	**65.65±1.27**	**67.81±1.03**	39.55±1.95	**79.91±1.61**	**92.32±0.48**	**95.36±0.60**	**91.51±0.55**	**94.24±0.43**

the increase in specificity is substantially larger than the decrease in sensitivity. Together with the simultaneous improvements in ACC and AUC, this suggests that the reduction in false-positive predictions outweighs the loss in sensitivity, resulting in improved overall classification performance.

In the second set of experiments, we compare BioBGG with methods other than basic correlation-based graphs. Since these models typically incorporate their own dedicated brain graph generation modules, directly replacing them with the proposed method would be inappropriate. Instead, we pair a common analysis model (AE) with BioBGG to assess its effectiveness relative to other generation approaches. The results are presented in Tab. II. For advanced static brain graphs, the proposed method shows clear advantages. Specifically, it significantly outperforms sparsified correlation-based graphs (e.g., IBGNN and IGS) as well as the task- and temporal-order–aware model FBNETGEN on the ABIDE and ADHD datasets. On the HCP dataset, FBNETGEN achieves performance comparable to BioBGG, likely due to the higher quality of fMRI data in this dataset. These results further demonstrate the robustness and generalizability of the proposed method across diverse data types. For dynamic brain graph–based methods, only ACIFBN achieves a competitive result on the ADHD dataset. Overall, however, their performance is generally inferior to that of static methods and consistently lower than that achieved by the proposed model.

In summary, the proposed brain graph generation method demonstrates robust and consistent improvements across different datasets and evaluation metrics. It not only enhances performance when integrated with common analysis models but also outperforms advanced static and dynamic baselines. These results highlight both the effectiveness and the broad adaptability of the proposed method.

F. Ablation Study (Q2)

In this section, we first evaluate the contribution of each component by comparing the performance of BioBGG against its ablated variants, where the temporal max pooling module, the critical ROI enhancement loss, or both are removed.

As shown in Tab. III, temporal max pooling provides the most substantial performance gain across all three baselines on all datasets. Without this component, the model averages similarity scores across all time points to estimate the presence of co-activation patterns. Consequently, it fails to capture transient co-activation patterns and loses the advantage of frame-wise co-activation pattern detection. These results highlight the importance of capturing transient co-activation patterns and demonstrate the effectiveness of the temporal max pooling strategy.

Regarding the critical ROI enhancement loss, its contribution is not substantial when applied in isolation. This is because its role is partly to encourage the max pooling operation to identify signal patterns with high amplitudes in critical ROIs; when used alone, this function is restricted. However, as shown in Tab. III, when combined with the max pooling operation, it consistently boosts model performance.

The above analysis demonstrates that temporal max pooling is the key factor of performance improvements, as it enables the model to capture transient co-activation patterns that correlation-based methods overlook. While the critical ROI enhancement loss alone contributes little, its combination with temporal max pooling further enhances performance, confirming the complementary roles of the two components.

TABLE III
PERFORMANCE COMPARISON (MEAN ± STD, %) OF BIOBGG WITH AE, GCN, AND BNT UNDER DIFFERENT ABLATION SETTINGS. THE HIGHEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Model	Component	ABIDE				ADHD				HCP			
		Accuracy	AUC	Sensitivity	Specificity	Accuracy	AUC	Sensitivity	Specificity	Accuracy	AUC	Sensitivity	Specificity
AE	None	64.02±1.58	70.27±0.81	71.32±1.08	58.67±1.01	61.81±1.62	64.04±0.78	34.22±2.55	80.31±3.43	83.15±1.82	88.87±1.45	82.41±1.72	87.81±1.45
	w/o Maxp.	64.24±1.51	70.84±0.72	72.26±1.25	58.23±1.21	61.92±1.54	64.24±1.04	34.87±2.54	80.14±2.77	84.48±1.62	90.21±1.32	82.89±1.15	91.01±0.98
	w/o $\mathcal{L}_{ROI}$	70.24±1.24	77.43±1.02	76.33±0.94	63.86±1.52	64.94±1.82	67.78±1.24	32.81±2.21	**85.81±1.66**	90.41±1.45	95.13±0.56	88.42±1.02	92.23±0.85
	BioBGG	**71.23±1.58**	**77.56±0.83**	**76.52±1.31**	**65.49±1.82**	**65.65±1.27**	**67.81±1.03**	**39.55±1.95**	79.91±1.61	**92.32±0.48**	**95.36±0.60**	**91.51±0.55**	**94.24±0.43**
GCN	None	56.87±1.27	64.87±1.43	**94.64±3.48**	11.20±2.56	60.57±1.57	62.54±1.23	23.10±3.41	84.83±3.94	80.66±2.07	86.67±1.66	80.12±2.32	86.24±2.52
	w/o Maxp.	64.53±1.01	69.37±0.88	75.52±1.07	52.92±2.28	62.87±1.47	64.87±1.03	27.89±2.95	81.86±4.03	81.33±1.97	87.64±1.47	81.30±1.87	88.66±1.21
	w/o $\mathcal{L}_{ROI}$	66.56±1.58	76.01±1.01	75.83±1.77	54.43±3.95	63.78±1.28	65.85±1.21	27.67±4.31	**86.27±2.85**	87.58±1.57	92.46±1.22	85.94±2.07	90.08±1.56
	BioBGG	**69.48±0.57**	**76.65±0.97**	76.25±1.38	**63.91±0.62**	**64.52±1.04**	**66.61±1.37**	**41.85±1.73**	77.56±1.42	**90.58±1.35**	**95.67±0.52**	**89.48±1.45**	**92.13±1.48**
BNT	None	64.44±1.35	70.28±0.88	70.05±1.28	57.99±1.91	61.47±1.48	64.31±0.99	36.59±1.92	83.61±2.72	84.77±1.61	89.83±1.15	82.68±1.43	91.38±1.08
	w/o Maxp.	64.42±0.98	70.31±0.99	69.59±1.62	58.61±1.73	61.34±1.32	64.15±1.12	36.47±1.92	83.45±2.84	84.95±1.41	90.08±0.98	83.58±1.44	91.18±1.10
	w/o $\mathcal{L}_{ROI}$	69.23±0.82	74.05±1.01	**75.24±1.81**	62.74±1.64	65.18±1.31	66.15±1.22	38.53±2.74	**83.72±1.24**	90.27±0.82	95.34±0.24	88.58±1.07	92.48±1.12
	BioBGG	**70.15±1.27**	**74.40±1.88**	73.38±1.76	**68.04±2.81**	**65.88±1.25**	**66.64±1.13**	**41.69±3.68**	81.95±3.88	**92.64±0.72**	**95.80±0.58**	**92.10±1.30**	**94.03±1.07**

TABLE IV
CLASSIFICATION ACCURACY (MEAN ± STD, %) USING JOINTLY TRAINED AND FROZEN BIOBGG GRAPH GENERATORS. FROZEN-AE/GCN/BNT INDICATE GENERATORS PRETRAINED WITH AE, GCN, AND BNT, RESPECTIVELY. THE HIGHEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Generator Source	ABIDE			ADHD			HCP		
	AE	GCN	BNT	AE	GCN	BNT	AE	GCN	BNT
Joint Training	71.23±1.58	69.48±0.57	70.15±1.27	65.65±1.27	64.52±1.04	65.88±1.25	92.32±0.48	90.58±1.35	**92.64±0.72**
Frozen-AE	70.72±0.83	69.62±0.43	69.83±0.92	**66.14±0.88**	**64.73±1.12**	**66.23±1.05**	**92.56±0.53**	90.38±1.15	92.50±0.66
Frozen-GCN	**71.32±0.95**	**69.86±0.46**	70.34±0.95	65.49±0.84	64.60±1.02	65.71±1.34	92.20±0.60	90.72±1.22	92.44±0.78
Frozen-BNT	70.94±0.97	69.29±0.57	**70.42±0.86**	65.73±0.99	64.43±1.17	66.10±1.01	91.96±0.64	**90.84±1.21**	92.59±0.59

In addition to the contribution of individual components, we further investigated whether the performance improvements originate primarily from the joint optimization of the graph generator and classifier or from the connectivity patterns learned by BioBGG itself. To this end, we first jointly trained BioBGG with a downstream classifier and then froze the learned graph generator. The resulting pretrained generator was subsequently combined with different downstream classifiers without further optimization of the generator.

As shown in Table IV, graph generators pretrained together with AE, GCN, and BNT were each evaluated using all three classifiers. The performance differences between jointly trained and frozen generators are generally small. These results suggest that the observed performance gains do not result from end-to-end joint optimization. Instead, BioBGG is able to learn meaningful and transferable connectivity patterns that generalize across different classifier architectures.

G. Hyperparameter Analysis (Q3)

In this section, we analyze the impact of key hyperparameters on model performance. Three brain graph analysis baselines are included, each representing a distinct model type. The hyperparameters examined include the window size w_s of the max pooling layer, the weight α of the critical ROI enhancement loss, the kernel size k of the candidate patterns, and the number of candidate patterns n. Although the optimal settings are obtained through hyperparameter search, the results show that strong performance can be achieved across a wide range of configurations. This indicates that the model's effectiveness is not merely the result of parameter tuning but reflects its inherent robustness.

1) Impact of The Max pooling Window Size: The max pooling operation in the proposed model is responsible for selecting signal patterns that best match the candidate patterns, thereby indicating the presence of those patterns in the brain. Increasing the window size helps capture less frequently occurring transient co-activation patterns, which explains the performance improvements observed as the window size increases, particularly when it is below 20. However, as shown in Fig. 3, performance declines once the window size exceeds 35 in the ABIDE and ADHD datasets. This is because many samples in these datasets have very limited sequence lengths; a larger window size reduces the number of frames available for estimating the existence score. When the window size is greater than 35, many samples are left with only one frame from a single window to assess the presence of a co-activation pattern, which significantly decreases reliability. In contrast, the HCP dataset contains samples with signal sequences of length 2,000, window sizes ranging from 25 to 200 all yield strong performance. This suggests that the impact of a large window size reflects the properties of the data rather than limitations of the model itself. Based on these findings, we set the window size to 25 to ensure that each sample provides at least two frames for existence score estimation, given that most real-world fMRI data have a length exceeding 50.

2) Impact of the Critical ROIs Enhancement Loss Weight α: The critical ROI enhancement restriction amplifies the difference between the top-k elements and the remaining ones. When the weight α is set to zero, the learned key patterns fail to concentrate on critical ROIs, resulting in poor interpretability and degraded task performance. As shown in Fig. 4, introducing the critical ROI enhancement loss consistently improves model performance. However, a slight performance

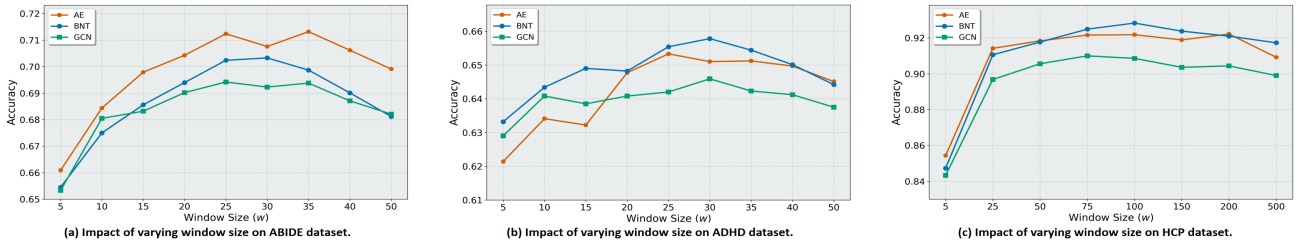


Fig. 3. Impact of varying window sizes on ABIDE, ADHD and HCP datasets.

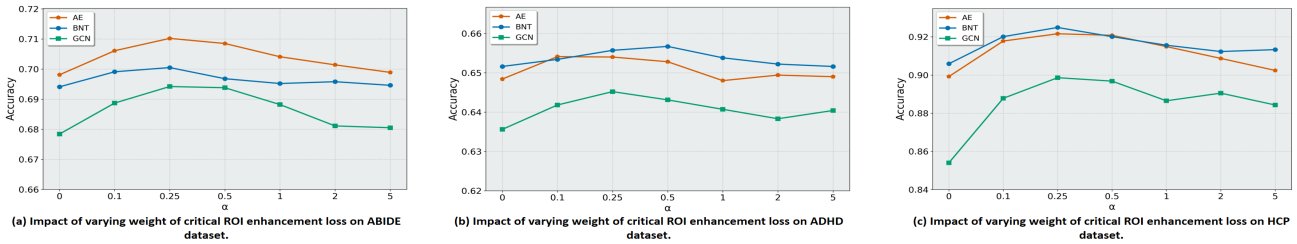


Fig. 4. Impact of varying the value of α on ABIDE, ADHD and HCP datasets.

decline can be observed when α exceeds 0.25, likely due to conflicts between this objective and the primary optimization objective. Notably, even when $\alpha > 0.25$, the overall performance remains higher than that achieved without this restriction, further demonstrating its robustness and positive contribution to model performance.

3) Impact of the Number of Candidate Patterns and Their Kernel Size: In BioBGG, the number of candidate patterns and the kernel size jointly determine the number of connections in the final graph. In general, increasing the number of measured connections leads to better performance. As shown in Fig. 5, increasing the number of candidate patterns consistently boosts performance across different classifiers and datasets, but at the cost of higher computational complexity. To balance effectiveness and efficiency, we set the number of candidate patterns to 800, as further increases yield only marginal improvements.

Regarding the kernel size, as discussed in Section IV-A4, a smaller kernel size provides a more accurate connection estimation. Although increasing the kernel size allows more connections to be measured, it does not necessarily lead to better performance. As illustrated in Fig. 5, when the number of candidate patterns is small, a larger kernel size tends to yield better results because it captures more connections. However, as the number of candidate patterns increases, the quality of the estimated connections becomes the dominant factor. In this case, performance with smaller kernels improves rapidly, whereas larger kernels (e.g., $k = 5$) become less effective. Since we set the number of candidate patterns at 800 to control computational cost, kernel sizes of 3 or 4 achieve comparable performance. Considering the greater reliability of connections estimated with smaller kernels, we set the kernel size to 3.

In summary, the same hyperparameter setting, including 800 candidate patterns, a kernel size of 3, and a max pooling window size of 25, is adopted across all three datasets despite their differences in sample size, disease type, and data distribution. The consistently strong performance obtained under this unified setting, together with the hyperparameter analysis presented above, suggests that these hyperparameters are robust and can serve as reasonable default choices for new brain graph analysis tasks.

*H. Time Cost Analysis (**Q4**)*

As discussed in Section IV-D, the proposed BioBGG has a time complexity of $\mathcal{O}(N \cdot n \cdot L)$, while correlation-based methods have a complexity of $\mathcal{O}(N^2 \cdot L)$. In this study, with $n = 800$ and $N = 200$, the proposed method is slightly more computationally intensive than the correlation-based methods, which is consistent with the results shown in Tab. V.

Compared with advanced static methods, the proposed approach achieves significantly lower computational cost. Specifically, IGS and IBGNN demand considerable computation to derive connection masks during training, whereas FBNETGEN employs recurrent neural networks on fMRI signal sequences to capture temporal dependencies, further increasing the cost in both training and inference.

For dynamic brain graph methods, a sequence of graphs is constructed using sliding window, which substantially increases the computational cost of both graph generation and subsequent analysis, leading to more than ten times the training and inference time compared with the proposed method.

In summary, the time complexity of the proposed BioBGG is comparable to that of correlation-based graphs, more efficient than advanced static graph generation methods, and substantially more efficient than dynamic approaches.

*I. Reliability and Interpretability Analysis (**Q5**)*

In this section, we evaluate the ability of BioBGG to identify biomarkers and assess its reliability compared with correlation-based connectivity estimation methods.

1) Identifying Critical Connections for Disorder Diagnosis: In BioBGG, the importance of a connection in predicting a brain disorder is quantified by the weights of the ROIs from the candidate patterns associated with that connection. The

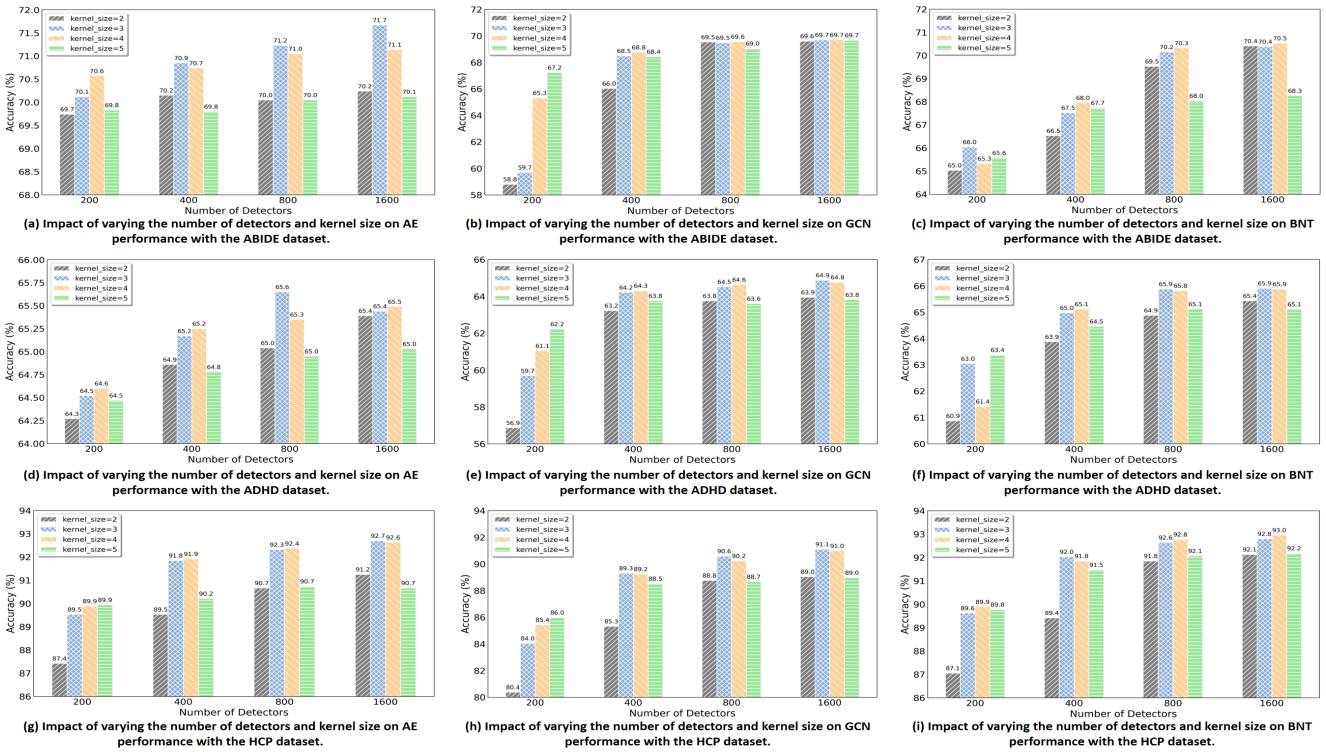


Fig. 5. Impact of varying number of candidate patterns and their kernel size.

TABLE V
RUNTIME COMPARISON DURING TRAINING AND INFERENCE ON THE ABIDE DATASET: AVERAGED TIME PER SAMPLE (MS). "CORR. STATIC" DENOTES TEMPORAL CORRELATION–BASED STATIC BRAIN GRAPHS.

Network Type	Model	Training	Inference
Corr. Static	AE	0.11	0.06
	GCN	0.34	0.30
	BNT	0.24	0.18
BioBGG	AE	0.28	0.20
	GCN	0.35	0.32
	BNT	0.36	0.33
Advanced Static	IGS	33.27	0.30
	IBGNN	1.48	0.33
	FBNETGEN	0.96	0.84
Dynamic	STAGIN	6.41	4.87
	STIGR	20.51	17.08
	ACIFBN	6.06	4.75

overall importance score matrix, denoted as $\mathbf{A}_{\text{imp}}$, is computed as $\mathbf{A}_{\text{imp}} = \frac{\mathbf{P}\mathbf{P}^{\top}}{|\mathbf{P}'\mathbf{P}'^{\top}|}$. Based on these importance scores, we select the five most critical connections to evaluate the interpretability of the proposed model. For ASD diagnosis, the top five connections are illustrated in Fig. 6 (a). We then conduct a group-level connection analysis of these five connections, with the results shown in Fig. 6 (b). In this analysis, taller green bars indicate decreased connectivity, while shorter green bars indicate increased connectivity. From this analysis, we identify decreased connectivity in ASD patients involving the Thalamus and Cuneus, the bilateral Cingulate Gyri, the Right Cerebellum and Left Postcentral Gyrus, the Left Temporal Gyrus and Left Precuneus, and the Right Middle Frontal Gyrus and Superior Frontal Gyrus. These findings are consistent with prior studies [45], [46], [47], [48]. For ADHD diagnosis, as shown in Fig. 7 (a) and (b), we identify decreased connectivity within subdivisions of the Middle Frontal Gyrus, between the Middle Frontal Gyrus and the Inferior Frontal Gyrus, and within cerebellar subdivisions, consistent with previous studies [49], [50]. In addition, we observed increased connectivity in the Precentral Gyrus and the Supramarginal Gyrus, in agreement with earlier reports [51].

2) Reliability Comparison Between Connectivity Estimation with BioBGG and Correlation: We evaluate the reliability of BioBGG from two perspectives: its stability in capturing inter-group connectivity discrepancies and its ability to detect transient connection states overlooked by temporal correlation.

For the first aspect, Panel (b) of both Fig. 6 and Fig. 7 presents group-level differences in connection strength between patients and HCs for the top five connections. In both figures, the left column shows correlation-based results and the right column shows BioBGG-based results; the upper rows correspond to the training set, while the lower rows correspond to the test set. In the training set, both correlation-based and BioBGG-estimated connections exhibit similar inter-group discrepancies. However, these discrepancies are notably more stable across training and test sets when estimated by BioBGG. For example, in Fig. 6 (b), correlation-estimated connections between the ROI-78 and the ROI-104 show decreased connectivity in patients relative to HCs in the training set but shift to over-connectivity in the test set. Similarly, connections between the ROI-95 and the ROI-85 show increased connectivity in the training set but shift to decreased connectivity in

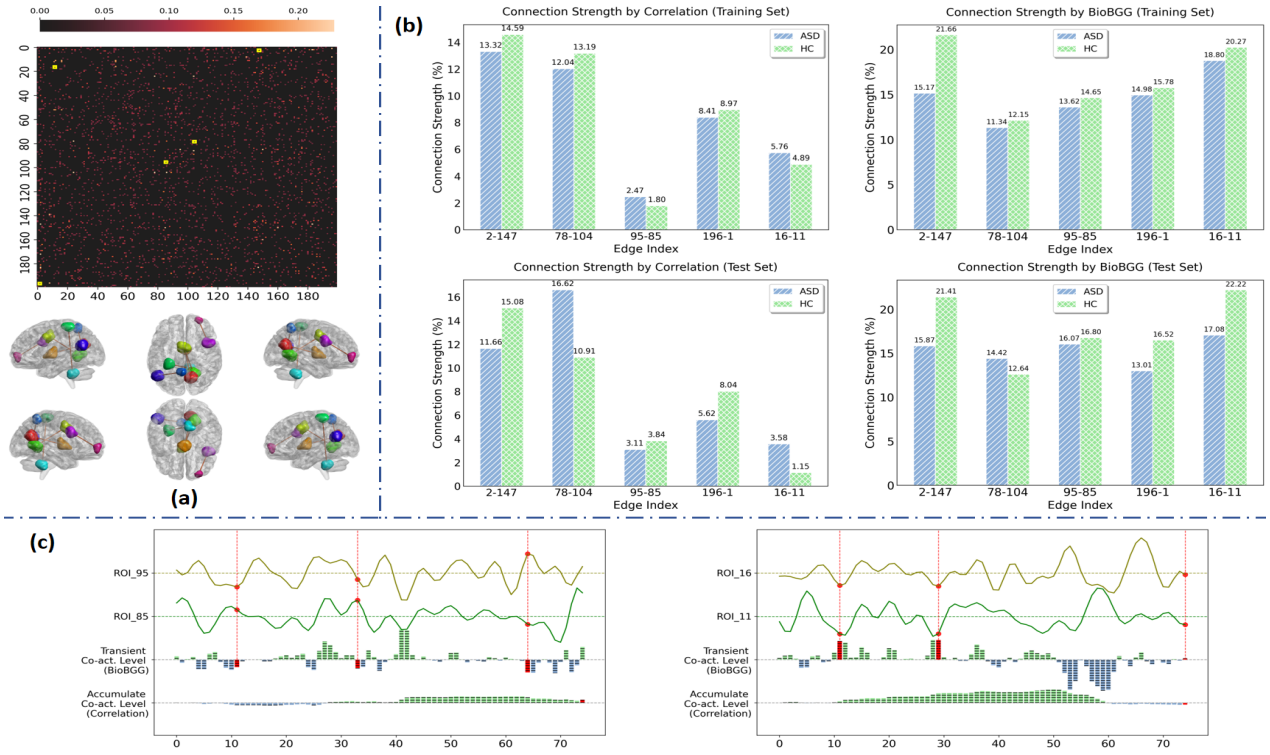


Fig. 6. Interpretability analysis on the ABIDE dataset. (a) Saliency map of connection importance, with the top five connections highlighted in yellow boxes, alongside their corresponding visualization in the brain. (b) Group-level comparison of the top five important connections estimated by correlation- and BioBGG-based methods. (c) Examples showing that BioBGG captures transient co-activation patterns that temporal correlation methods fail to detect.

the test set. Comparable shifts are observed in Fig. 7 (b) for connections between the ROI-189 and the ROI-76, between the ROI-148 and the ROI-41, and between the ROI-122 and the ROI-14. By contrast, BioBGG produces more stable results, with only one inconsistent case observed in Fig. 6(b) for the connection between the ROI-78 and the ROI-104. This reliable detection of inter-group differences helps generate stable results. Such stability may arise from BioBGG's ability to consistently capture transient connection states that are sometimes overlooked by correlation-based estimation, which we further demonstrate in the second aspect below.

For the second aspect, we select samples where correlation-based methods produce different estimates from BioBGG among the top five important connections. Detailed explanations are provided in Fig. 6 (c) and Fig. 7 (c). Each figure contains four rows: the first two rows display fMRI signal sequences from two ROIs; the third row depicts transient co-activation levels, with positive values shown in green, negative values in blue, and frames selected for connection estimation by BioBGG are highlighted in red; and the fourth row illustrates the correlation calculation process, where accumulated co-activation levels across the entire sequence yield a final correlation estimate, depicted as the height of the last bar in this row. From these figures, it is evident that both positive and negative transient co-activations exist for the same pair of ROIs. Temporal correlation–based estimation aggregates these transient states, leading to cancellation of positive and negative contributions and ultimately produces a near-zero connection strength, overlooking transient neuro-interactions. By contrast, BioBGG performs frame-wise analysis, successfully capturing transient connections. This enables more reliable brain graph generation and contributes to improved performance.

VI. Limitation and Discussion

One main limitation of the proposed brain graph generation method is that it does not explicitly model temporal order information. While the method emphasizes the presence of co-activation patterns in the brain, it does not account for their temporal order. This limitation is also shared by most existing brain graph construction methods, which estimate functional connectivity from the entire fMRI sequence. Several studies have recognized this drawback and attempted to model temporal order using recurrent neural networks or transformers [13], [43], [44], achieving competitive performance. However, these works rarely evaluate whether their models truly benefit from incorporating temporal order information. To directly assess this, we selected three temporal order–aware baselines, namely FBNETGEN [43], STAGIN [13], and STIGR [44], and compared their performance on brain graphs constructed from both original and temporally shuffled signal sequences. In the shuffled experiments, the time points of the fMRI sequence were randomly permuted, with the signals associated with each time point relocated accordingly. This random shuffling was applied consistently across both training and testing. If temporal order were critical and the models effectively captured it, shuffling should have led to substantially degraded performance. Surprisingly, the results in Tab. VI show that shuffling the temporal order of fMRI signals had little effect on the performance of the baselines across all three datasets.

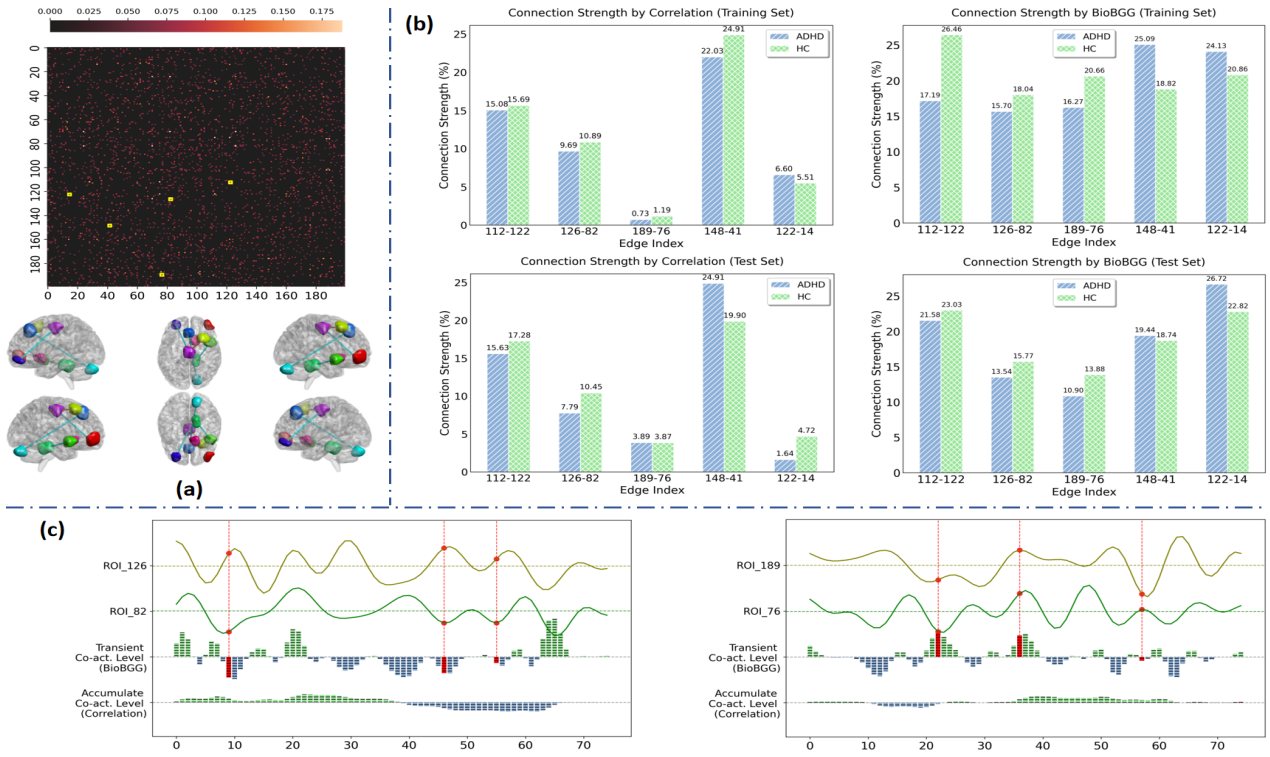


Fig. 7. Interpretability analysis on the ADHD dataset. (a) Saliency map of connection importance, with the top five connections highlighted in yellow boxes, alongside their corresponding visualization in the brain. (b) Group-level comparison of the top five important connections estimated by correlation- and BioBGG-based methods. (c) Examples showing that BioBGG captures transient co-activation patterns that temporal correlation methods fail to detect.

TABLE VI
CLASSIFICATION ACCURACY (MEAN ± STD, %) OF TEMPORAL ORDER–AWARE MODELS WITH ORIGINAL AND SHUFFLED SEQUENCES.

	ABIDE			**ADHD**			**HCP**		
	FBNETGEN	STAGIN	STIGR	FBNETGEN	STAGIN	STIGR	FBNETGEN	STAGIN	STIGR
Original Sequence	66.27 ± 1.54	56.57 ± 1.01	$\mathbf{61.32 \pm 1.84}$	$\mathbf{61.20 \pm 1.30}$	54.25 ± 1.25	60.22 ± 1.47	$\mathbf{91.14 \pm 0.74}$	$\mathbf{88.46 \pm 4.55}$	$\mathbf{84.56 \pm 2.32}$
Shuffled Sequence	$\mathbf{66.48 \pm 1.35}$	$\mathbf{57.43 \pm 1.24}$	60.47 ± 1.33	60.78 ± 1.41	$\mathbf{56.28 \pm 1.52}$	$\mathbf{61.12 \pm 1.28}$	91.02 ± 0.99	87.67 ± 4.12	82.35 ± 3.42

These findings raise an important open question: how can temporal information be reliably captured, and does doing so improve brain disorder diagnosis? In future work, we will investigate this question and explore methods for leveraging temporal features that may enhance diagnostic performance.

VII. CONCLUSION

In this study, we addressed the limitations of traditional temporal correlation–based brain graph generation methods, which often fail to capture transient co-activation patterns occurring over short time scales. To overcome this challenge, we proposed BioBGG, which performs frame-wise analysis of fMRI signals to extract temporally localized co-activation patterns and leverages these patterns to generate brain graphs. Extensive experiments demonstrate that the proposed method achieves superior predictive performance, maintains high computational efficiency, and is seamlessly compatible with common brain graph analysis models and datasets. Furthermore, interpretability analysis confirms the reliability of BioBGG in identifying meaningful biomarkers for brain disorder diagnosis. These findings underscore the potential of frame-level connection estimation to advance brain graph analysis. In future work, we plan to enhance this approach by incorporating temporal information of co-activation patterns present in fMRI data to further improve its diagnostic performance and the interpretability of the generated brain graphs.

REFERENCES

[1] J. V. Hull, L. B. Dokovna, Z. J. Jacokes, C. M. Torgerson, A. Irimia, and J. D. Van Horn, "Resting-state functional connectivity in autism spectrum disorders: a review," *Frontiers in Psychiatry*, vol. 7, p. 205, 2017.
[2] H. Cui, W. Dai, Y. Zhu, X. Li, L. He, and C. Yang, "Interpretable graph neural networks for connectome-based brain disorder analysis," in *MICCAI*. Springer, 2022, pp. 375–385.
[3] X. Kan, W. Dai, H. Cui, Z. Zhang, Y. Guo, and C. Yang, "Brain network transformer," *NeurIPS*, vol. 35, pp. 25 586–25 599, 2022.
[4] G. Li, M. Duda, X. Zhang, D. Koutra, and Y. Yan, "Interpretable sparsification of brain graphs: Better practices and effective designs for graph neural networks," in *SIGKDD*, 2023, pp. 1223–1234.
[5] A. Bannadabhavi, S. Lee, W. Deng, R. Ying, and X. Li, "Community-aware transformer for autism prediction in fmri connectome," in *MICCAI*. Springer, 2023, pp. 287–297.
[6] X. Luo, J. Wu, J. Yang, H. Chen, Z. Li, H. Peng, and C. Zhou, "Knowledge distillation guided interpretable brain subgraph neural networks for brain disorder exploration," *TNNLS*, vol. 36, no. 2, pp. 3559–3572, 2024.

[7] Y. Gu, S. Peng, Y. Li, L. Gao, and Y. Dong, "Fc-hgnn: A heterogeneous graph neural network based on brain functional connectivity for mental disorder identification," *Information Fusion*, vol. 113, p. 102619, 2025.

[8] B. Thapaliya, E. Akbas, J. Chen, R. Sapkota, B. Ray, P. Suresh, V. D. Calhoun, and J. Liu, "Brain networks and intelligence: A graph neural network based approach to resting state fmri data," *Medical Image Analysis*, vol. 101, p. 103433, 2025.

[9] T. Meindl, S. Teipel, R. Elmouden, S. Mueller, W. Koch, O. Dietrich, U. Coates, M. Reiser, and C. Glaser, "Test–retest reproducibility of the default-mode network in healthy individuals," *Human Brain Mapping*, vol. 31, no. 2, pp. 237–246, 2010.

[10] Z. Shehzad, A. C. Kelly, P. T. Reiss, D. G. Gee, K. Gotimer, L. Q. Uddin, S. H. Lee, D. S. Margulies, A. K. Roy, B. B. Biswal *et al.*, "The resting brain: unconstrained yet reliable," *Cerebral Cortex*, vol. 19, no. 10, pp. 2209–2229, 2009.

[11] X. Liu, N. Zhang, C. Chang, and J. H. Duyn, "Co-activation patterns in resting-state fmri signals," *Neuroimage*, vol. 180, pp. 485–494, 2018.

[12] L. Li, Q. Zheng, Y. Xue, M. Bai, and Y. Mu, "Coactivation pattern analysis reveals altered whole-brain functional transient dynamics in autism spectrum disorder," *European Child & Adolescent Psychiatry*, vol. 33, no. 12, pp. 4313–4324, 2024.

[13] B.-H. Kim, J. C. Ye, and J.-J. Kim, "Learning dynamic graph representation of brain connectome with spatio-temporal attention," *NeurIPS*, vol. 34, pp. 4314–4327, 2021.

[14] Z. Guan, J. Yu, Z. Shi, X. Liu, R. Yu, T. Lai, C. Yang, H. Dong, R. Chen, and L. Wei, "Dynamic graph transformer network via dual-view connectivity for autism spectrum disorder identification," *Computers in Biology and Medicine*, vol. 174, p. 108415, 2024.

[15] L. Liu, G. Wen, P. Cao, T. Hong, J. Yang, X. Zhang, and O. R. Zaiane, "Braintgl: A dynamic graph representation learning model for brain network analysis," *Computers in Biology and Medicine*, vol. 153, p. 106521, 2023.

[16] R. Liu, Y. Hu, J. Wu, K.-C. Wong, Z.-A. Huang, Y.-A. Huang, and K. Chen Tan, "Dynamic graph representation learning for spatio-temporal neuroimaging analysis," *IEEE Transactions on Cybernetics*, vol. 55, no. 3, pp. 1121–1134, 2025.

[17] H. Cui, W. Dai, Y. Zhu, X. Kan, A. A. C. Gu, J. Lukemire, L. Zhan, L. He, Y. Guo, and C. Yang, "Braingb: a benchmark for brain network analysis with graph neural networks," *TMI*, vol. 42, no. 2, pp. 493–506, 2022.

[18] A. Said, R. Bayrak, T. Derr, M. Shabbir, D. Moyer, C. Chang, and X. Koutsoukos, "Neurograph: Benchmarks for graph machine learning in brain connectomics," *NeurIPS*, vol. 36, pp. 6509–6531, 2023.

[19] A. S. Heinsfeld, A. R. Franco, R. C. Craddock, A. Buchweitz, and F. Meneguzzi, "Identification of autism spectrum disorder using deep learning and the abide dataset," *NeuroImage: Clinical*, vol. 17, pp. 16–23, 2018.

[20] F. Almuqhim and F. Saeed, "Asd-saenet: a sparse autoencoder, and deep-neural network model for detecting autism spectrum disorder (asd) using fmri data," *Frontiers in Computational Neuroscience*, vol. 15, p. 654315, 2021.

[21] X. Li, Y. Zhou, N. Dvornek, M. Zhang, S. Gao, J. Zhuang, D. Scheinost, L. H. Staib, P. Ventola, and J. S. Duncan, "Braingnn: Interpretable brain graph neural network for fmri analysis," *Medical Image Analysis*, vol. 74, p. 102233, 2021.

[22] G. Wen, P. Cao, H. Bao, W. Yang, T. Zheng, and O. Zaiane, "Mvs-gcn: A prior brain structure learning-guided multi-view graph convolution network for autism spectrum disorder diagnosis," *Computers in Biology and Medicine*, vol. 142, p. 105239, 2022.

[23] S. Yu, S. Jin, M. Li, T. Sarwar, and F. Xia, "Long-range brain graph transformer," *NeurIPS*, vol. 37, pp. 24 472–24 495, 2024.

[24] Z. Yin and K. Yue, "Temporal resonant graph network for representation learning on dynamic graphs," *Applied Intelligence*, vol. 53, no. 7, pp. 7466–7483, 2023.

[25] Q. Zhu, S. Li, X. Meng, Q. Xu, Z. Zhang, W. Shao, and D. Zhang, "Spatio-temporal graph hubness propagation model for dynamic brain network classification," *TMI*, vol. 43, no. 6, pp. 2381–2394, 2024.

[26] F. Zhao and E. E. Kuruoglu, "Sequential monte carlo graph convolutional network for dynamic brain connectivity," in *ICASSP*. IEEE, 2024, pp. 7535–7539.

[27] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" *ICLR*, 2019.

[28] K. Lin, B. Jie, P. Dong, X. Ding, W. Bian, and M. Liu, "Convolutional recurrent neural network for dynamic functional mri analysis and brain disease identification," *Frontiers in Neuroscience*, vol. 16, p. 933660, 2022.

[29] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[30] S. Gadgil, Q. Zhao, A. Pfefferbaum, E. V. Sullivan, E. Adeli, and K. M. Pohl, "Spatio-temporal graph convolution for resting-state fmri analysis," in *MICCAI*. Springer, 2020, pp. 528–538.

[31] J. Wang, N. C. Dvornek, P. Duan, L. H. Staib, P. Ventola, and J. S. Duncan, "Stnagnn: Data-driven spatio-temporal brain connectivity beyond fc," *arXiv preprint arXiv:2406.12065*, 2024.

[32] J. Zhang, X. Wu, X. Tang, L. Zhou, L. Wang, W. Wu, and D. Shen, "Asynchronous functional brain network construction with spatiotemporal transformer for mci classification," *TMI*, 2024.

[33] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *AISTATS*. PMLR, 2010, pp. 249–256.

[34] R. M. Hutchison, T. Womelsdorf, E. A. Allen, P. A. Bandettini, V. D. Calhoun, M. Corbetta, S. Della Penna, J. H. Duyn, G. H. Glover, J. Gonzalez-Castillo *et al.*, "Dynamic functional connectivity: promise, issues, and interpretations," *Neuroimage*, vol. 80, pp. 360–378, 2013.

[35] C. Craddock, Y. Benhajali, C. Chu, F. Chouinard, A. Evans, A. Jakab, B. S. Khundrakpam, J. D. Lewis, Q. Li, M. Milham *et al.*, "The neuro bureau preprocessing initiative: open sharing of preprocessed neuroimaging data and derivatives," *Frontiers in Neuroinformatics*, vol. 7, no. 27, p. 5, 2013.

[36] P. Bellec, C. Chu, F. Chouinard-Decorte, Y. Benhajali, D. S. Margulies, and R. C. Craddock, "The neuro bureau adhd-200 preprocessed repository," *Neuroimage*, vol. 144, pp. 275–286, 2017.

[37] D. C. Van Essen, S. M. Smith, D. M. Barch, T. E. Behrens, E. Yacoub, K. Ugurbil, W.-M. H. Consortium *et al.*, "The wu-minn human connectome project: an overview," *Neuroimage*, vol. 80, pp. 62–79, 2013.

[38] C. Craddock, S. Sikka, B. Cheung, R. Khanuja, S. S. Ghosh, C. Yan, Q. Li, D. Lurie, J. Vogelstein, R. Burns *et al.*, "Towards automated analysis of connectomes: The configurable pipeline for the analysis of connectomes (c-pac)," *Front Neuroinform*, vol. 42, no. 10.3389, 2013.

[39] R. C. Craddock, G. A. James, P. E. Holtzheimer III, X. P. Hu, and H. S. Mayberg, "A whole brain fmri atlas generated via spatially constrained spectral clustering," *Human Brain Mapping*, vol. 33, no. 8, pp. 1914–1928, 2012.

[40] S. M. Smith, C. F. Beckmann, J. Andersson, E. J. Auerbach, J. Bijsterbosch, G. Douaud, E. Duff, D. A. Feinberg, L. Griffanti, M. P. Harms *et al.*, "Resting-state fmri in the human connectome project," *Neuroimage*, vol. 80, pp. 144–168, 2013.

[41] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *ICLR*, 2015.

[42] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *ICLR*, 2017.

[43] X. Kan, H. Cui, J. Lukemire, Y. Guo, and C. Yang, "Fbnetgen: Task-aware gnn-based fmri analysis via functional brain network generation," in *MIDL*. PMLR, 2022, pp. 618–637.

[44] R. Liu, Y. Hu, J. Wu, K.-C. Wong, Z.-A. Huang, Y.-A. Huang, and K. C. Tan, "Dynamic graph representation learning for spatio-temporal neuroimaging analysis," *IEEE Transactions on Cybernetics*, 2025.

[45] R. Ayub, K. L. Sun, R. E. Flores, V. T. Lam, B. Jo, M. Saggar, and L. K. Fung, "Thalamocortical connectivity is associated with autism symptoms in high-functioning adults with autism and typically developing adults," *Translational psychiatry*, vol. 11, no. 1, p. 93, 2021.

[46] Y. Zhou, L. Shi, X. Cui, S. Wang, and X. Luo, "Functional connectivity of the caudal anterior cingulate cortex is decreased in autism," *PloS one*, vol. 11, no. 3, p. e0151879, 2016.

[47] H. Wang, Y. Liu, and Y. Ding, "Identifying diagnostic biomarkers for autism spectrum disorder from higher-order interactions using the ped algorithm," *Neuroinformatics*, vol. 22, no. 3, pp. 285–296, 2024.

[48] E. T. Rolls, Y. Zhou, W. Cheng, M. Gilson, G. Deco, and J. Feng, "Effective connectivity in autism," *Autism Research*, vol. 13, no. 1, pp. 32–44, 2020.

[49] Y. Wang, L. Ma, J. Wang, Y. Ding, W. Men, S. Tan, J.-H. Gao, S. Qin, Y. He, Q. Dong *et al.*, "Connections between the middle frontal gyrus and the dorsoventral attention network are associated with the development of attentional symptoms," *Biological Psychiatry*, vol. 97, no. 5, pp. 531–539, 2025.

[50] M. Cundari, S. Vestberg, P. Gustafsson, S. Gorcenco, and A. Rasmussen, "Neurocognitive and cerebellar function in adhd, autism and spinocerebellar ataxia," *Frontiers in Systems Neuroscience*, vol. 17, p. 1168666, 2023.

[51] H. Zhang, Y. Zhao, W. Cao, D. Cui, Q. Jiao, W. Lu, H. Li, and J. Qiu, "Aberrant functional connectivity in resting state networks of adhd patients revealed by independent component analysis," *BMC Neuroscience*, vol. 21, no. 1, p. 39, 2020.